INCH-POUND

MIL-C-48483C(AR)
09 May 1995
(see 6.4)

MILITARY SPECIFICATION

CIRCUIT CARD ASSEMBLY - COMPARATOR, TYPE I

This specification is approved for use by the U.S. Army Armament, Research, Development and Engineering Center (ARDEC) and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements and quality assurance provisions for the Circuit Card Assembly Comparator Type I, 11732530 (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-F-13926 - Fire Control Materiel, General Specification Governing the Manufacture and Inspection of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use to improving this document should be addressed to: Commander, U.S. Army ARDEC, ATTN: AMSTA-AR-EDE-S, Picatinny Arsenal, New Jersey 07806-5000 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1240

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

MIL-I-45607 - Inspection Equipment; Acquisition, Maintenance, and Disposition of

STANDARDS

MILITARY

MIL-STD-100 - Engineering Drawing Practices
MIL-STD-454 - Standard General Requirements for Electronic Equipment
MIL-STD-810 - Environmental Test Methods
MIL-STD-2000 - Standard Requirements for Soldered Electrical and Electronic Assemblies
MIL-STD-2073-1 - Procedures for Development and Application of Packaging Requirements

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from: DODSSP - Customer Service, Standardization Documents Order Desk, 700 Robbins Avenue, Bldg 4D, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issue shall be those in effect on date of the solicitation.

U.S. Army Armament Research Development and Engineering Center (ARDEC)

DRAWINGS

11732530 - Circuit Card Assembly - Comparator, Type I

(Copies of Government drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets or MS standards), the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained (See control provisions for additional precedence criteria).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Fabrication. The Comparator, Type I circuit card assembly, herein referred to as the assembly, shall be manufactured in accordance with Drawing 11732530 and drawings pertaining thereto and, when assembled, shall meet the requirements of this specification (see 4.5.1).

3.1.1 Function. The assembly shall provide the following:

a. Compare the computer power, ammo select, - epsilon and eta signals for correct, predetermined values and provide logic for decoding results of the comparisons.

b. General logic signals necessary for illuminating either a card malfunction lamp or a computer OK lamp.

3.1.2 General specifications. (see 4.6.3)

3.1.2.1 Manufacture and inspection. The following provisions of MIL-F-13926 apply:

a. Dimensions and tolerances
b. Effect of protective coating on dimension (Inorganic type coatings)
c. Part identification and marking
d. Workmanship (including applicable portions of MIL-STD-2000 and MIL-STD-454, Requirement 9)

3.1.2.2 Standards of manufacture.

3.1.2.2.1 Assembly and soldering. The requirements of MIL-STD-2000 shall apply, as a minimum.

3.1.2.2.2 Interchangeability. The assembly shall be manufactured in accordance with the interchangeability requirements as specified in MIL-STD-100.

3.1.3 Ambient conditions. Standard ambient conditions shall be as follows:

a.	Temperature	73° ± 18°F
b.	Relative humidity	50 percent + 30 percent
c.	Atmospheric pressure	28.5 + 2.0 -3.0 in. Hg.

3.2 First article. When specified in the contract or purchase order (see 6.2), the contractor shall furnish sample units for first article inspection and approval (see 4.4).

3.3 Performance. Unless otherwise specified, the assembly shall meet the performance requirements specified herein under standard ambient conditions of 3.1.3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

3.3.1 Computer A1 and A2 cards fail comparator logic. (see 4.6.1)

3.3.1.1 Phase 1. With the power sources (items 2.1, 2.2, 2.4 and 2.6 of Table I), lamp loads (items 1.1 and 1.2 of Table I), and interconnections (item 4 of Table I) applied to the assembly; and with the items input conditions of Table II established in any item number order, the corresponding lamp illumination conditions shall be specified in Table II.

3.3.1.2 Phase 2. With the power sources (items 2.1, 2.2, 2.4 and 2.6 of Table I), lamp loads (item 1.1 of Table I), and interconnections (item 4 of Table I) applied to the assembly and with pins B7 and A23 connected to ground pins A3, B3, the signal source (item 3.2 of Table I) applied and adjusted to +12.469 $\pm$ 0.005V rms phase 2, and with the signal source (item 3.1 of Table I) applied and:

a. With item 3.1 of Table I voltage adjusted to the nominal center point +2.023 $\pm$ 0.005V dc, lamp 1.1 shall extinguish.

b. With item 3.1 of Table I voltage increased from the nominal voltage, lamp 1.1 shall illuminate. This change shall occur with item 3.1 voltage adjusted within the outward bound upper limit +2.243 to +2.354V dc.

c. With item 3.1 of Table I voltage decreased from the outward bound upper limit, lamp 1.1 shall extinguish. This change shall occur with item 3.1 voltage adjusted within the inward bound upper limit range +2.221 to 2.325V dc.

d. With item 3.1 of Table I voltage decreased from the inward bound upper limit, lamp 1.1 shall illuminate. This change shall occur with item 3.1 voltage adjusted within the outward bound lower limit range +1.698 to +1.805V dc.

e. With item 3.1 of Table I voltage increased, lamp 1.1 shall extinguish. This change shall occur with item 3.1 voltage adjusted within the inward bound lower limit range +1.727 to +1.828V dc.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

TABLE I. Loads, power, signals and interconnections.

Item	Conditions	Characteristics	Conditions
1.0	Loads		Connected between the following pins:
1.1	Lamp	24V lamp in series with 24V dc (Item 2.1)	A12(-) and 24V dc (see Item 2.1)
1.2	Lamp	24V lamp in series with 24V dc (Item 2.1)	A15(-) and 24V dc (see Item 2.1)
2.0	Power sources		Applied between pin(s)
2.1	24V dc	Tolerance: ±2V dc Ripple: 200 mv p-p max Current: 500 ma min	B16(-) and load lamps Items 1.1 and 1.2 (+)
2.2	15V dc	Tolerance: ±0.30V dc Ripple: 20 mv p-p max Current: 500 ma min	(A1, B1) (+) and (A3, B3) (-), and between pins as specified
2.3	10V dc	Tolerance: ±0.010V dc Ripple: 20 mv p-p max Current: 500 ma min	A22(+) and (A3, B3) (-), and between pins as specified
2.4	5.250V dc	Tolerance: ±0.15V dc or as otherwise specified Ripple: 20 mv p-p max Current: 500 ma min	(A30, B30) (+) and A13, B13 (-), and between pins as specified
2.5	-10V dc	Tolerance: ± 0.010V dc Ripple: 20 mv p-p max Current: 500 ma min	A17(-) and (A3, B3) (+), and between pins as specified
2.6	-15V dc	Tolerance: ±0.030V dc Ripple: 20 mv p-p max Current: 500 ma min	A4(-) and (A3, B3) (+), and between pins as specified
3.0	Signal sources		When specified:
3.1	Selectable: 0 to 7V dc	Tolerance as specified Ripple: 20 mv p-p max Current: 500 ma min	Applied to pin A5(+) with return to pins A3, B3
3.2	Selectable: 0 to 17V dc	Tolerance as specified Ripple: 20 mv p-p max Current: 500 ma min	Applied to pin B9(+) with return to pins (A3,B3)
3.3	Selectable: -6 to -14V dc	Tolerance as specified Ripple: 20 mv p-p max Current: 500 ma min	Applied to pin B6(-) with return to pins (A3, B3)
3.4	Selectable: 0 to 5V rms 400 Hz	Tolerances: ±0.005V rms, or as otherwise specified; ±1 Hz Phase: 2	Applied to pin A29 with return to pins (A3, B3)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

TABLE I. Loads, power, signals and interconnections. (Cont)

Item	Conditions	Characteristics	Conditions
3.5	Selectable: 0 to 5V rms,	Tolerances: ±0.005V rms; ±1 Hz Phase: 1 (180° out-of-phase with phase 2 in item 3.4)	Applied to pin A28 with return to pins (A3, B3)
3.6	Selectable: 0 to 5V rms 400 Hz	Tolerances: ±0.005V rms, or as otherwise specified; ±1 Hz Phase: 2	Applied to pins B29 with return to pins (A3, B3)
3.7	Selectable: 0 to 5V rms, 400 Hz	Tolerances: ±0.005V rms; ±1 Hz Phase: 1 (180° out-of-phase with phase 2 in item 3.6)	Applied to pin B26 with return to pins (A3, B3)
3.8	Selectable: 0 to 9V rms 400 Hz	Tolerances: ±0.005V rms, or otherwise specified; ±1 Hz Phase: 2	Applied to pin A2 with return to pins (A3, B3)
3.9	30V P-P, 400 Hz (MR1)	Square wave Tolerances: ±0.50V; ±1 Hz Phase: 1	Applied to pin A20 with return to pins (A3, B3)
3.10	30V P-P, 400 Hz (MR2)	Square wave Tolerances: ±0.50V; ±1 Hz Phase: 2 (180° out-of-phase with phase 1 in Item 3.9)	Applied to pin B20 with return to pins (A3, B3)
4.0	Interconnections		Interconnections of the following pins:
	---	---	Pin A5 with Pins A6, B5 B4, and B2
	---	---	Pins A3, B3, with pins A13, B13, and B8

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

TABLE II. Computer A1 and A2 cards fail comparator logic.

Input Conditions at Pins						Illumination Condition at Load Lamps, Table I, Items 1/	
Item	2/ B7	2/ A23	(V dc) A5 3/	(V dc) B9 4/	(V rms) A2 5/	Pin 1.1 A12	Pin 1.2 A15
1	0	1	2.023 ± 0.005	12.469 ± 0.005	5.000 ± 0.005	N/A 6/	X
2	0	1	2.023 ± 0.005	12.469 ± 0.005	1.000 max	-	-
3	0	1	1.698 max	12.469 ± 0.005	5.000 ± 0.005	X	-
4	1	N/A	1.698 max	12.469 ± 0.005	N/A	-	N/A
5	0	0	2.023 ± 0.005	12.469 ± 0.005	5.000 ± 0.005	N/A	-
6	0	1	1.698 max	11.529 max	5.000 ± 0.005	-	X
7	1	1	1.698 max	11.529 max	5.000 ± 0.005	N/A	-

1/ X = illuminated; - = not illuminated
2/ 1 = 5.25 ± 0.15 V dc, 0 = GND interconnection with Pins A3, B3
3/ Table I, Item 3.1 signal
4/ Table I, Item 3.2 signal
5/ Table I, Item 3.8 signal
6/ N/A = indicates not applicable condition

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

3.3.1.3 Phase 3. With the power sources (items 2.1, 2.2, 2.4 and 2.6 of Table I), lamp load (item 1.2 of Table I), and interconnections (item 4 of Table I) applied to the assembly and with pins B7 and A23 connected to ground pins A3, B3, item 3.1 of Table I applied and adjusted to +2.023 ± 0.005V dc, item 3.8 of Table I applied and adjusted to 5.000 ± 0.005V rms phase 2, and with item 3.2 of Table I applied and:

a. With item 3.2 of Table I voltage adjusted to the nominal center point of +12.496 ± 0.005V dc, lamp 1.2 shall be extinguished.

b. With item 3.2 of Table I voltage increased from the nominal voltage, lamp 1.2 shall be illuminated. This change shall occur with item 3.2 voltage adjusted within the outward bound upper limit range +13.102 to +13.439V dc.

c. With item 3.2 of Table I voltage decreased from the outward bound upper limit, lamp 1.2 shall be extinguished. This change shall occur with voltage adjusted within the inward bound upper limit range +13.043 to +13.361V dc.

d. With item 3.2 of Table I voltage decreased from the inward bound upper limit, lamp 1.2 shall be illuminated. This change shall occur with item 3.2 voltage adjusted within the outward bound lower limit range +11.529 to +11.835V dc.

e. With item 3.2 of Table I voltage increased from the outward bound lower limit, lamp 1.2 shall be extinguished. This change shall occur with item 3.2 voltage adjusted within the inward bound lower limit range +11.604 to +11.895V dc.

3.3.2 Computer -10V comparator logic. With the power sources (items 2.2, 2.3, 2.4 and 2.6 of Table I), interconnections (item 4 of Table I), and the signal source (item 3.8 of Table I) applied to the assembly and adjusted to 5.000 ± 0.005V rms phase 2 and: (see 4.6.1.2)

a. With the input signal source item 3.3 of Table I applied at the value -10.000 ± 0.010V dc, nominal voltage, the output level at pin B27 shall be at logic level one, 5.0 +0.5/-2.6V dc, measured with respect to pins A13, B13.

b. With a decrease in the input signal item 3.3 of Table I from the nominal voltage, the output level at pin B27 shall change to logic level zero, 0.0 ± 0.5V dc. The input signal voltage item 3.3 of Table I shall fall within the outward bound lower limit range -10.201 ± 0.049V dc.

c. With an increase in the input signal voltage item 3.3 of Table I from the outward bound lower limit, the output

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

Purchase Order, Shipments Date

level at pin B27 shall change to logic level one, 5.0 +0.5/-2.6 V dc. The input signal voltage item 3.3 of Table I shall fall within the inward bound lower limit range -10.182 ± 0.047V dc.

d. With an increase in the input signal voltage item 3.3 of Table I from the inward bound lower limit, the output level at pin B27 shall change to logic level zero, 0.0 +0.5/-2.6V dc. The input signal voltage item 3.3 of Table I shall fall within the outward bound upper limit range -9.797 ± 0.049V dc.

e. With a decrease in the input signal voltage item 3.3 of Table I from the outward bound upper limit, the output level at pin B27 shall change to logic level one, 5.0 +0.5/-2.6V dc. The input signal voltage item 3.3 of Table I shall fall within the inward bound upper limit range -9.816 ± 0.047V dc.

3.3.3 AC Ref 2 (comp) comparator. With the power sources (items 2.2, 2.3, 2.4 and 2.6 of Table I), interconnections (item 4 of Table I), and the signal source (item 3.3 of Table I) applied to the assembly and adjusted to -10.000 ± 0.010V dc and: (see 4.6.1.3)

a. With the signal source item 3.8 of Table I applied at the value +5.000 ± 0.005V rms phase 2, the output level at pin B27 shall be at logic level one, 5.0 +0.5/-2.6V dc, measured with respect to pins A13, B13, 5.0.

b. With the signal source item 3.8 of Table I applied at the value +9.000 ± 0.005V rms phase 2, the output level at pin B27 shall be at logic level zero, 0.0 ± 0.5V dc.

c. With the signal source item 3.8 of Table I applied at the value +1.000 ± 0.005V rms phase 2, the output level at pin B27 shall be at logic level zero, 0.00 + 0.500V dc.

3.3.4 Ammo select comparator logic. With power sources (items 2.2 thru 2.6 of Table I) and interconnections (item 4 of Table I) applied to the assembly, and with the input conditions of Table III established in any item number order, the corresponding output logic levels shall be as specified therein (see 4.6.1.4).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

TABLE III. Ammo select comparator logic.

Item	Input Conditions at Pins							Output at Pins 6/		
	A19 1/	A7	A5 2/	B9 3	B6 4/	A2 5/	B24 7/	A10	B11	B22
1	NC	GND	2.023	12.469	-10.000	5.000	NA 8/	1	1	NA
2	NC	GND	1.698 9/	12.469	-10.000	5.000	NA	0	NA	NA
3	-15V dc	GND	1.698 9/	12.469	-10.000	5.000	NA	NA	1	NA
4	NC	GND	2.023	11.529 9/	-10.000	5.000	0	NA	1	0
5	-15V dc	GND	2.023	11.529 9/	-10.000	5.000	0	NA	1	0
6	NC	GND	2.023	12.469	-10.000	1.000 9/	NA	0	NA	NA
7	-15V dc	GND	2.023	12.469	-10.000	1.000 9/	NA	NA	1	NA
8	NC	GND	2.023	12.469	- 7.000	5.000	NA	0	NA	NA
9	-15V dc	GND	2.023	12.469	- 7.000	5.000	NA	NA	1	NA
10	-15V dc	GND	2.023	12.469	-10.000	5.000	0	0	0	1
11	-15V dc	GND	2.023	12.469	-10.000	5.000	1	NA	0	0
12	-15V dc	-15V dc	2.023	12.469	-10.000	5.000	0	1	1	NA

1/ NC = no connection, open;
-15V dc = -15.000 ± 0.010V dc (Table I, Item 2.6 power)
2/ ± 0.005V dc (Table I, Item 3.1 signal)
3/ ± 0.005V dc (Table I, Item 3.2 signal)
4/ ± 0.005V dc (Table I, Item 3.3 signal)
5/ ± 0.005V rms (Table I, Item 3.8 signal)
6/ Logic 1 = +5.0 +0.5/-2.6V dc
Logic 0 = 0.0 ± 0.5V dc
7/ 1 = +5.25 ± 0.15V dc; 0 = 0.0 ± 0.5V dc, interconnection with pins A3, B3
8/ NA = not an applicable condition
9/ Maximum. Voltage tolerance is not applicable.

Source: https://assist.dla.mil -- Downloaded: 2026-02-26T03:56Z
Check the source to verify that this is the current version before use.

3.3.5 -Epsilon comparator logic. With the power sources (item 2.2 thru 2.6 of Table I), signal sources (items 3.4, 3.9 and 3.10 of Table I) and interconnections (item 4 of Table I) applied to the assembly, with the input conditions of Table IV established in any item number order and: (see 4.6.1.5)

a. With the signal source item 3.4 of Table I adjusted to the nominal voltage, the output level at pin B12 shall be a logic level one, +5.0 +0.5/-2.6V dc, measured with respect to pins A3, B3.

b. With a decrease in the signal source item 3.4 of Table I, the output level at pin B12 shall change to a logic level zero, 0.0 ± 0.5V dc. The signal source voltage item 3.4 shall fall within the outward bound lower limit range.

c. With an increase in the signal source item 3.4 of Table I, the output level at pin B12 shall change to a logic level one, 5.0 +0.5/-2.6V dc. The signal source voltage item 3.4 shall fall within the inward bound lower limit range.

d. With an increase in the signal source item 3.4 of Table I from the inward bound lower limit, the output level at pin B12 shall change to a logic level zero, 0.0 ± 0.5V dc. The signal source voltage item 3.4 shall fall within the outward bound upper limit range.

e. With a decrease in the signal source item 3.4 of Table I, the output level at pin B12 shall change to a logic level one, 5.0 +0.5/-2.6V dc. The signal source voltage item 3.4 shall fall within the inward bound upper limit range.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

TABLE IV. -Epsilon comparator logic initial conditions.

Item	Input Conditions at Pins						Input Condition at Pin A29 2/				
	A28 1/	A26 3/	A27 3/	A25 3/	B23 3/	B25 3/	Outward Bound Lower Limit	Inward Bound Lower Limit	Inward Bound Upper Limit	Outward Bound Upper Limit	Nominal Voltage
1	4.000 ±0.005 V rms	GND	GND	-15V dc	-15V dc	-15V dc	0.391 to 0.408 V rms	0.402 to 0.417 V rms	0.472 to 0.488 V rms	0.482 to 0.499 V rms	0.444 ±0.005 V rms
2	4.000 ±0.005 V rms	GND	-15V dc	-15V dc	-15V dc	-15V dc	0.421 to 0.438 V rms	0.431 to 0.488 V rms	0.502 to 0.518 V rms	0.512 to 0.530 V rms	0.474 ±0.005 V rms
3	4.000 ±0.005 V rms	-15V dc	-15V dc	-15V dc	GND	-15V dc	0.364 to 0.381 V rms	0.374 to 0.390 V rms	0.445 to 0.461 V rms	0.455 to 0.472 V rms	0.416 ±0.005 V rms
4	4.000 ±0.005 V rms	-15V dc	-15V dc	GND	-15V dc	-15V dc	2.227 to 2.302 V rms	2.287 to 2.311 V rms	2.358 to 2.382 V rms	2.368 to 2.393 V rms	2.333 ±0.005 V rms
5	4.000 ±0.005 V rms	-15V dc	-15V dc	-15V dc	-15V dc	GND	0.882 to 0.901 V rms	0.892 to 0.911 V rms	0.963 to 0.981 V rms	0.973 to 0.993 V rms	0.936 ±0.005 V rms

1/ Table I, Item 3.5 signal
2/ Table I, Item 3.4 signal
3/ GND = 0.0 ±0.5V dc, Interconnection with pins A3, B3;
-15V dc = -15.000 ±0.030V dc power, Item 2.6 of Table I.

Source: https://assist.dla.mil -- Downloaded: 2026-02-26T03:56Z
Check the source to verify that this is the current version before use.

3.3.6 Eta comparator logic. With the power sources (items 2.2 thru 2.6 of Table I), signal sources (items 3.6, 3.9 and 3.10 of Table I)and interconnections (item 4 of Table I) applied to the assembly, with the input conditions of Table V established in any item number order and: (see 4.6.1.6)

a. With the signal source item 3.6 of Table I adjusted to the nominal voltage, the output level at pin B18 shall be logic level one, 5.0 +0.5/-2.6V dc, measured with respect to pins A3, B3.

b. With a decrease in the signal source item 3.6 of Table I, the output level at pin B18 shall change to logic level zero, 0.0 ± 0.5V dc. The signal source voltage item 3.6 shall fall within the outward bound lower limit range.

c. With an increase in the signal source item 3.6 of Table I, the output level at pin B18 shall change to logic level one, +5.0 +0.5/-2.6V dc. The signal source voltage item 3.6 shall fall within the inward bound lower limit range.

d. With an increase in the signal source item 3.6 of Table I from the inward bound lower limit, the output level at pin B18 shall change to logic level zero, 0.0 ± 0.5V dc. The signal source voltage item 3.6 shall fall within the outward bound upper limit range.

e. With a decrease in the signal source item 3.6 of Table I, the output level at pin B18 shall change to logic level one, 5.0 +0.5/-2.6V dc. The signal source voltage item 3.6 shall fall within the inward bound upper limit range.

3.3.7 Comparator decoding logic. With the power sources (items 2.2 thru 2.6 of Table I), signal sources (items 3.9 and 3.10 of Table I) and the interconnections set up as specified in Table I, and with pin A25 connected to pins A3, B3, and with the item input conditions of Table II established in any item number order, the corresponding output logic levels shall be as specified in Table VII (see 4.6.1.7).

3.3.8 D_L OK logic. With the power sources (item 2.4 of Table I) and the interconnections (item 4 of Table I) applied to the assembly, and with the input conditions of Table III established in any item number order, the corresponding output voltages at pin B17 shall be as specified in Table VIII (see 4.6.1.8).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

TABLE V. Eta comparator logic initial conditions.

Item	Input Conditions at Pins						Input Condition at Pin B29 2/				
	B26 1/	A26 3/	A27 3/	A25 3/	B25 3/	B23 3/	Outward Bound Lower Limit	Inward Bound Lower Limit	Inward Bound Upper Limit	Outward Bound Upper Limit	Nominal Voltage
1	4.000 ±0.005 V rms	GND	GND	-15V dc	-15V dc	-15V dc	0.361 to 0.377 V rms	0.371 to 0.386 V rms	0.440 to 0.459 V rms	0.450 to 0.470 V rms	0.413 ±0.005 V rms
2	4.000 ±0.005 V rms	GND	-15V dc	-15V dc	-15V dc	-15V dc	0.391 to 0.407 V rms	0.401 to 0.416 V rms	0.470 to 0.489 V rms	0.479 to 0.500 V rms	0.443 ±0.005 V rms
3	4.000 ±0.005 V rms	-15V dc	-15V dc	-15V dc	-15V dc	GND	0.335 to 0.352 V rms	0.345 to 0.361 V rms	0.414 to 0.433 V rms	0.424 to 0.445 V rms	0.387 ±0.005 V rms
4	4.000 ±0.005 V rms	-15V dc	-15V dc	GND	-15V dc	-15V dc	2.245 to 2.269 V rms	2.255 to 2.278 V rms	2.324 to 2.351 V rms	2.334 to 2.363 V rms	2.301 ±0.005 V rms
5	4.000 ±0.005 V rms	-15V dc	-15V dc	-15V dc	-15V dc	GND	0.851 to 0.870 V rms	0.862 to 0.879 V rms	0.930 to 0.952 V rms	0.940 to 0.963 V rms	0.905 ±0.005 V rms

1/ Table I, Item 3.7 signal
2/ Table I, Item 3.6 signal
3/ GND = 0.0 ±0.5V dc, Interconneciton with pins A3, B3;
-15V dc = -15.000 ±0.030V dc power, Item 2.6 of Table I.

Source: https://assist.dla.mil -- Downloaded: 2026/0-26T03:56Z
Check the source to verify that this is the current version before use.

TABLE VI. Comparator decoding logic inputs.

Table I Item No.	Input Voltage
3.1	2.023 ± 0.005V dc
3.2	12.469 ± 0.005V dc
3.3	-10.000 ± 0.005V dc
3.5	4.000 ± 0.005V rms
3.7	4.000 ± 0.005V rms
3.8	5.000 ± 0.005V rms

TABLE VII. Comparator decoding logic.

	Input Conditions at Pins							Output at Pins 5/			
Item	A19 1/	A7 6/	A29 2/	B29 3/	B15 4/	B19 4/	B21 4/	B14	A16	A14	A21
1	-15V dc	NC	2.333	2.415	0	1	1	0	0	1	0
2	NC	NA	2.333	2.415	0	1	NA	0	1	0	NA
3	-15V dc	NC	2.446	2.415	NA	0	0	0	NA	0	0
4	NC	NC	2.333	2.302	NA	NA	1	0	0	NA	1
5	NC	NC	2.333	2.415	1	NA	0	NA	1	NA	1
6	-15V dc	NC	2.446	2.302	1	NA	NA	NA	1	NA	0
7	NC	NA	2.333	2.303	1	NA	NA	NA	1	NA	1
8	NC	NA	2.446	NA	NA	NA	NA	NA	NA	NA	1

1/ NC = No connection, open; -15V dc = -15.000 ± 0.030V dc (Table I, Item 2.6 power).
2/ ± 0.005V rms (Table I, Item 3.4 signal).
3/ ± 0.005V rms (Table I, Item 3.6 signal).
4/ 1 = +5.25 ± 0.15V dc; 0 = 0.0 ± 0.5V dc, interconnection with pins A3, B3.
5/ 1 = +5.0 +0.5/-2.6V dc; 0 = 0.0 ± 0.5V dc, NA = not applicable condition
6/ NA = Not an applicable condition; NC = No connection, open.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

TABLE VIII. D_L OK logic.

Item	Input Conditions at Pins 1/ B10	A9	Output at B17 2/
1	0	0	1
2	0	1	1
3	1	0	0
4	1	1	1

1/ 1 = 5.25 ± 0.15V dc (item 2.4 of Table I);
0 = 0.0 ± 0.5V dc (interconnection with pins A3, B3)
2/ 1 = 5.0 +0.5/-2.6V dc; 0 = 0.0 ± 0.5V dc

3.4 Environmental.

3.4.1 Temperature. The assembly shall be capable of operating as specified herein over the operating temperature range of +40°F to 100°F and shall not be damaged when subjected to the storage conditions ranging from -65°F to 160°F (see 4.6.2.1 and 4.6.2.2).

3.4.2 Vibration. The assembly shall be capable of operating as specified herein after withstanding exposure under the conditions of 3.1.3 to a sinusoidal vibration displacement of .03 inch double amplitude, 10 - 55 Hz in accordance with Figure 1. Duration of exposure shall be not less than 80 minutes in each of three mutually perpendicular axes (see 4.6.2.3).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

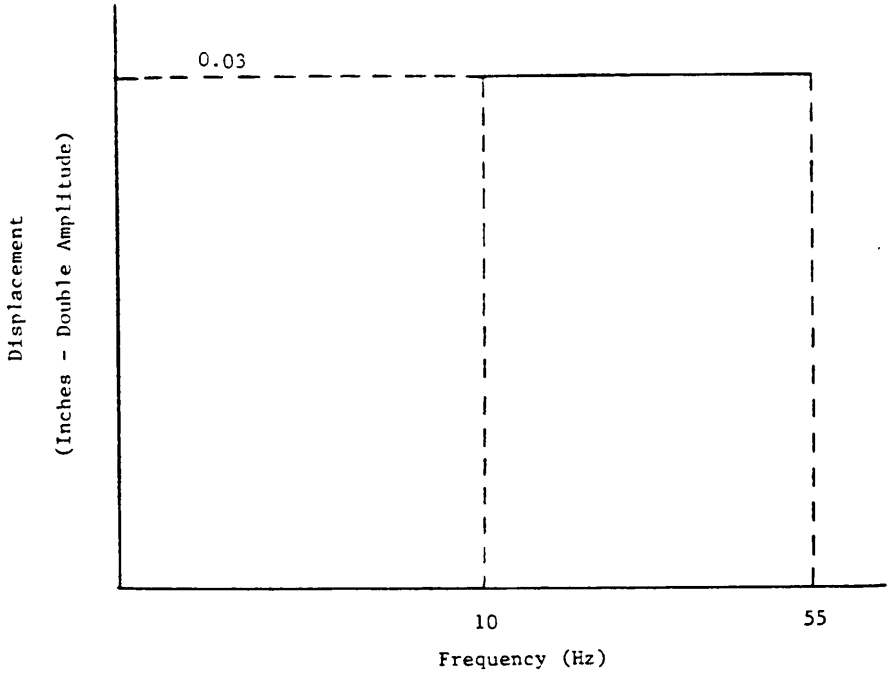


FIGURE 1. Vibration Curve.

3.4.3 Shock. The assembly shall be capable of operating as specified herein after withstanding exposure under the conditions of 3.1.3 to half sine wave shock pulses of 40 ± 4 gravity units (g's) for a duration of 18 ± 3 milliseconds (msec) in accordance with Figure 2. The assembly shall withstand three shock pulses applied in both directions along three mutually perpendicular axes (18 shocks). In addition, the assembly shall be capable of withstanding three half sine wave shock pulses of 100.0 ± 10.0 g's for a duration of 1.5 ± 0.2 msec applied in each direction along three mutually perpendicular axes (see 4.6.2.4).

A = 40 g's
D = 18 msec

A = 100 g's
D = 1.5 msec

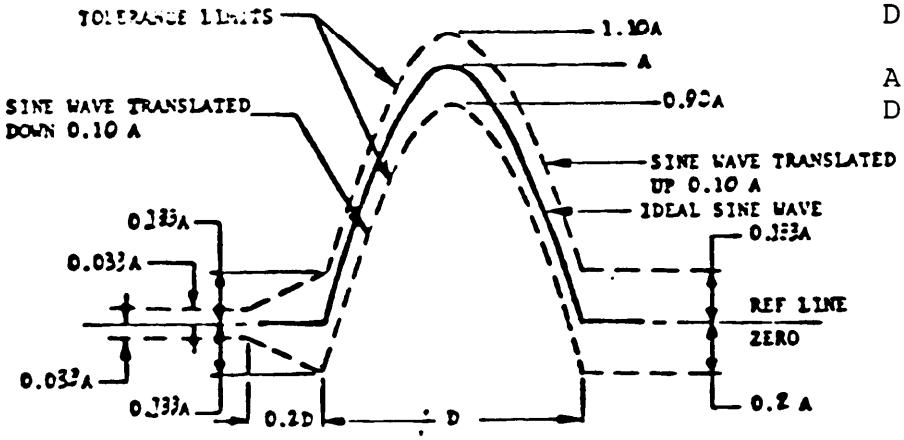


FIGURE 2. Shock Pulse.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

3.4.4 Temperature cycling. The assembly shall be capable of operating as specified herein after withstanding exposure to temperature cycling and soaking between the temperatures of -25°F and 140°F in accordance with Figure 3 (see 4.6.2.5).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in the specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Inspection equipment. Unless otherwise specified in the contract, the contractor shall supply, maintain and calibrate inspection equipment in accordance with the provisions of MIL-I-45607 and standard commercial practice. Contractor inspection equipment designs shall be submitted for Government approval as specified in the contract (see 6.5).

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)
b. Quality conformance inspection (see 4.5 and 6.8)

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the quality assurance provisions of MIL-F-13926 and the conditions of 3.1.3.

4.4 First article inspection. The requirements for first article approval and the designation of responsibility for first

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

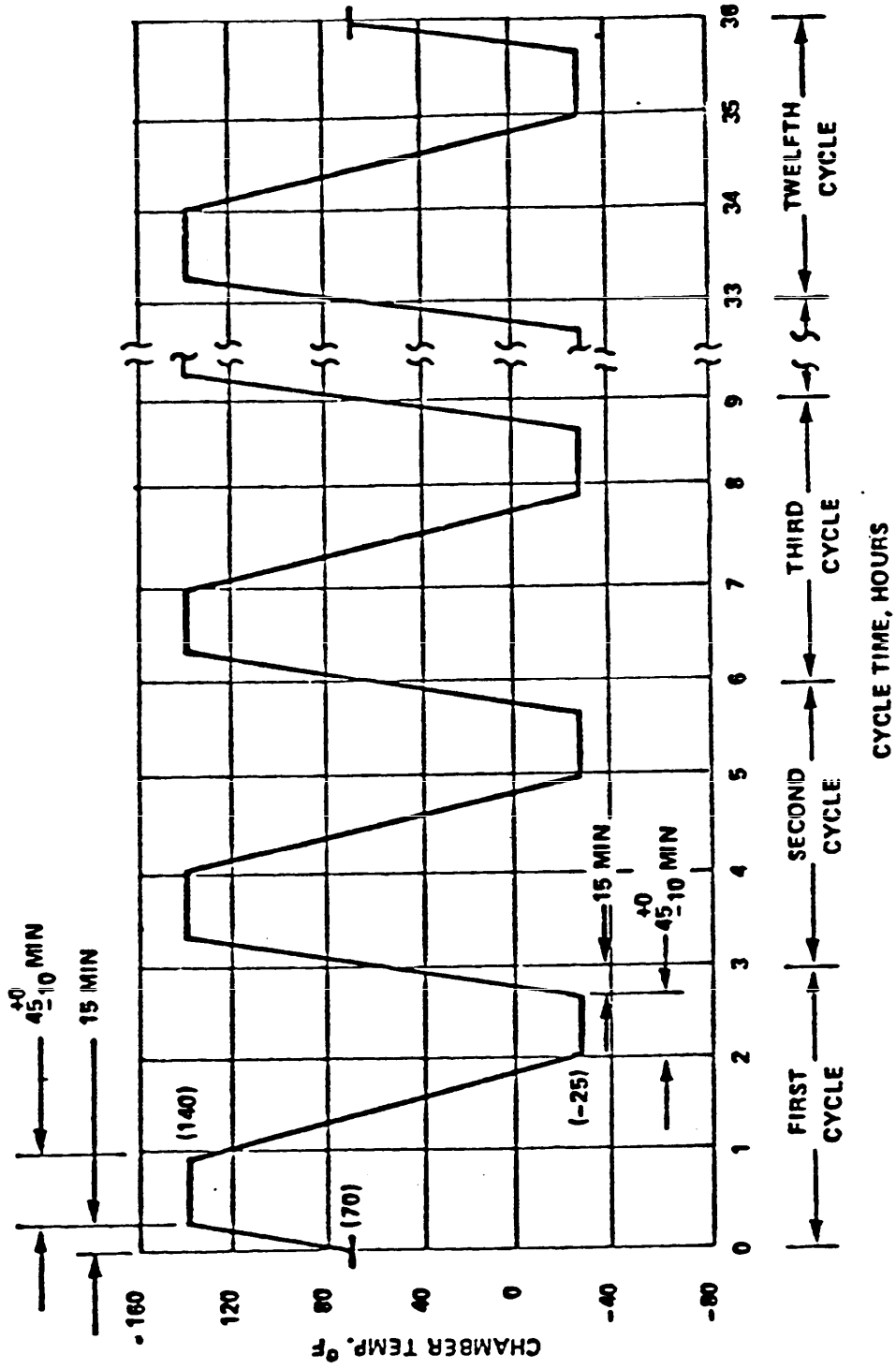


FIGURE 3. Temperature Profile

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

article inspection to either the Government or the contractor shall be specified in the contract (see 6.3).

4.4.1 Sample. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.4.2. The first article sample shall consist of two assemblies.

4.4.2 Inspections to be performed. Two assemblies shall be subjected to all the inspections specified in Tables IX, X and XI. Tests 301 and 302 of Table XI shall be performed first.

TABLE IX. Performance tests.

Item	Characteristics	Requirement	Test Procedure
101	Computer A1 and A2 cards fail comparator logic	3.3.1	4.6.1.1
102	Computer -10V comparator logic	3.3.2	4.6.1.2
103	Ac Ref 2 (comp) comparator	3.3.3	4.6.1.3
104	Ammo select comparator logic	3.3.4	4.6.1.4
105	-Epsilon comparator logic	3.3.5	4.6.1.5
106	Eta comparator logic	3.3.6	4.6.1.6
107	Comparator decoding logic	3.3.7	4.6.1.7
108	DL OK logic	3.3.8	4.6.1.8

TABLE X. Environmental tests.

Item	Characteristics	Requirement	Test Procedure
201	High Temperature	3.4.1	4.6.2.1
202	Low Temperature	3.4.1	4.6.2.2
203	Vibration	3.4.2	4.6.2.3
204	Shock	3.4.3	4.6.2.4
205	Temperature Cycling	3.4.4	4.6.2.5

TABLE XI. General tests.

Item	Characteristics	Requirement	Test Procedure
301	Fabrication	3.1	Applicable dwgs - Visual
302	General specifications	3.1.2	4.6.3 - Visual
303	Packaging inspection	5.1	4.6.4 - Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

4.4.3 Failure. Failure of any assembly to meet any requirement shall be cause for refusal to grant first article approval. The Government reserves the right to terminate first article inspection upon any failure of any assembly to comply with any stated requirement.

4.4.4 Responsibility. The contractor, whether or not responsible, shall inspect the sample for conformance to all contractual requirements and shall submit a record of this inspection and certificates of conformance for materials. The Government reserves the right to witness inspections performed by the contractor.

4.5 Quality conformance inspection.

4.5.1 Procurement conditions. The following inspection plans shall apply where the assembly is procured for use as a:

Use	Plan
a. Logistics spare assembly apart from the next higher assembly	A, B and C
b. Component assembly of the next higher assembly	B and C

4.5.2 Inspection plan A.

4.5.2.1 Environmental sample and tests. One assembly, as a control sample, shall be selected at random by the Government representative from each 50 assemblies produced or from each month's production, whichever occurs first. All the tests in Table X shall be applied separately.

4.5.2.2 Acceptance. When the sample assembly of the quantity produced fails to meet any specified requirement, the lot shall be rejected. Rejected lots shall be resubmitted for reinspection only after all assemblies are re-examined or retested and all defective assemblies are removed or defects corrected.

4.5.3 Inspection plan B.

4.5.3.1 Sample and tests. The sample shall be selected by the Government representative. All the tests in Table XI and only the test of item 205 in Table X shall apply. The following sampling plan (except that the visual examination for completeness, improper assembly and evidence of poor workmanship

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

per items 301 and 302 shall be performed 100%) shall be utilized:

100%, c=0 for lot sizes of 10 or less
n=10, c=0 for lot sizes of 11 to 150
n=32, c=0 for lot sizes greater than 150

where, n=sample size and c=0 requires zero defects during inspection for acceptance.

4.5.3.2 Acceptance. If the assembly fails to meet any specified requirement, the lot shall be rejected. Rejected lots shall be resubmitted as above.

4.5.4 Inspection plan C.

4.5.4.1 Sample and tests. Each assembly in every lot shall be subjected to all the tests in Table IX and shall be examined visually for completeness, improper assembly and evidence of poor workmanship per items 301 and 302 of Table XI. The conditions of 3.1.3 apply.

4.5.4.2 Acceptance. Where any one assembly fails to meet any specified requirement, the defective assembly shall be removed from the lot and resubmitted only after all defects have been corrected.

4.6 Methods of inspection.

4.6.1 Performance tests.

4.6.1.1 Computer A1 and A2 cards fail comparator logic. Apply the power sources, interconnections and other input conditions of 3.3.1. Verify the output conditions as specified in 3.3.1.

4.6.1.2 Computer -10V comparator logic. Apply the power sources, interconnections and other input conditions of 3.3.2. Verify the output conditions as specified in 3.3.2.

4.6.1.3 AC Ref 2 (comp) comparator. Apply the power sources, interconnections and other input conditions of 3.3.3. Verify the output conditions as specified in 3.3.3.

4.6.1.4 Ammo select comparator logic. Apply the power sources, interconnections and other input conditions of 3.3.4. Verify the logic output conditions as specified in 3.3.4

4.6.1.5 -Epsilon comparator logic. Apply the power sources, interconnections and other input conditions of 3.3.5. Verify the output conditions as specified in 3.3.5.

4.6.1.6 Eta comparator logic. Apply the power sources, interconnections and other input conditions of 3.3.6. Verify the

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

logic output conditions as specified in 3.3.6.

4.6.1.7 Comparator decoding logic. Apply the power sources, interconnections and other input conditions of 3.3.7. Verify the logic output conditions as specified in 3.3.7.

4.6.1.8 DL OK logic. Apply the power sources, interconnections and other input conditions of 3.3.8. Verify the logic output conditions as specified in 3.3.8.

4.6.2 Environmental tests.

4.6.2.1 High temperature. Subject the assembly to the high temperature test specified in MIL-STD-810, Method 501 except storage temperature shall be 160°F for 48 hours. At the conclusion of the storage temperature test, stabilize the assembly at 100°F and subject it to the performance tests of Table IX. After testing, return the assembly to room temperature and subject it to the performance tests of Table IX.

4.6.2.2 Low temperature. Subject the assembly to the low temperature test specified in MIL-STD-810, Method 502. Maintain the temperature at -65°F for a period of 12 hours minimum. At the conclusion of the storage temperature test, stabilize the assembly at 40°F and subject it to the performance tests of Table IX. After testing, return the assembly to room temperature and subject the assembly to the performance tests of Table IX.

4.6.2.3 Vibration. Vibration A as specified in 4.6.2.3.1 shall be conducted only for first article inspection. Vibration B as specified in 4.6.2.3.2 shall be conducted only for quality conformance inspection.

4.6.2.3.1 Vibration A. Mount the assembly in a suitable vibration fixture and subject the assembly to the vibration test specified in MIL-STD-810, Method 514 General Procedures, and 3.4.2. The vibration amplitude shall have a tolerance of ± 10%. The tolerance of the vibration frequency shall be ± 2%, or 1/2 Hz for frequencies below 25 Hz. Vibration sweeps shall be conducted with the druation of exposure not less than 80 minutes in each of three mutually perpendicular axes. At the conclusion of this test, subject the assembly to the performance tests of Table IX.

4.6.2.3.2 Vibration B. This test shall be conducted as in 4.6.2.3.1 except that the period of vibration shall be not less than 15 minutes in each axis.

4.6.2.4 Shock. Mount the assembly in a suitable shock fixture and subject it to the functional shock test for ground equipment specified in MIL-STD-810, Method 516, for a total of

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

three half sine wave shock pulses in both directions along three mutually perpendicular axes (18 shocks). Peak amplitude shall be 40 ± 4 g's with a time duration of 18 ± 3 msec in accordance with Figure 2. At the conclusion of this test, subject the assembly to the performance tests of Table IX. Repeat this test with peak amplitude of 100.0 ± 10.0 g's with a time duration of 1.5 ± 0.2 msec in accordance with Figure 2.

4.6.2.5 Temperature cycling. With power and loads of Table I applied, subject the assembly to the temperature profile specified in Figure 3. At the conclusion of this test, subject the assembly to the performance tests of Table IX.

4.6.3 General specifications. Verification shall be provided by examination, visually or through the use of standard measuring equipment, in accordance with 3.1.2.

4.6.4 Packaging inspection. The sampling and inspection of the preservation, packaging, packing and container marking shall be in accordance with MIL-STD-2073-1.

5. PACKAGING

5.1 Packaging, packing and marking. Packaging, packing and marking shall be in accordance with MIL-STD-2073-1. The level of protection shall be as specified in the procurement document (see 4.6.4 and 6.2d).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The Comparator, Type I circuit card (A7) assembly is intended for use in the Field Test Set, 11733200, which is used in the field for test of all of the M21 Computer subsystem components.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

c. Inspection provisions for first article (see 6.3).

d. Applicable levels of preservation, packaging, packing, and container marking (see 5.1 and 6.7).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

6.3 First article. The contracting officer should provide specific guidance to offerors for the submission of a first article sample to be tested as specified in 4.4.2. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Supersession. Previous revisions of MIL-C-48483(AR) were for the US Army Armament, Research, Development and Engineering Center's internal use only, and were not forwarded to the Naval Publications and Form Center for publication.

6.5 Submission of contractor inspection equipment designs for approval. Submit copies of designs as required to: Commander, U.S. Army ARDEC, ATTN: AMSTA-AR-QAC-F, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

6.6 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament, Research, Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as U.S. Army Armament, Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal. Technical data originally prepared by these activities is now under the cognizance of ARDEC.

6.7 Levels of packaging. The following definitions describe the three levels of packaging in accordance with MIL-STD-2073-1.

a. Level A - Adequate protection for indeterminate storage and worldwide distribution.

b. Level B - Adequate protection for known shipment conditions and storage conditions normally not exceeding one year.

c. Level C - Adequate protection for shipment to a known site for immediate use.

6.8 Alternative quality conformance provisions. Alternative quality conformance procedures, methods, or equivalent, such as statistical process control, tool control, other types of sampling procedures, etc., may be used by the contractor when they provide,

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

as a minimum, the level of quality assurance required by the provisions specified herein. Prior to applying such alternative procedures, methods, or equivalent, the contractor shall describe them in a written proposal submitted to the Government for evaluation. When required, the contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the specified quality assurance provision(s) herein. In cases of dispute as to whether the contractor's proposed alternative(s) provides equivalent assurance, the provisions of this specification shall apply. All approved alternative provisions shall be specifically incorporated into the contractor's quality program or detailed inspection system, as applicable. Unless otherwise specified in the contract, proposed alternative quality conformance provisions will be submitted by the contractor for evaluation by the technical activity.

6.9 Subject term (key word) listing.

Fire Control
Tank, M60A3
Test Set, Field, M21 Computer

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1240-0009)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.